The documentation and process conversion measures necessary to comply with this revision shall be completed by 27 September 1995

INCH-POUND

MIL-S-19500/486D
27 June 1995
SUPERSEDING
MIL-S-19500/486C
19 May 1991

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, COUPLER, OPTOELECTRIC, SOLID STATE, TYPES 4N22, 4N22A, 4N23, 4N23A, 4N24, 4N24A, 4N22U, 4N22AU, 4N23U, 4N23AU, 4N24U, 4N24AU, JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense

1 SCOPE

1 1 Scope This specification covers the detail requirements for solid state optically coupled isolators in which a gallium arsenide or gallium aluminum arsenide diode light source is optically coupled to a silicon NPN phototransistor Four levels of product assurance are provided for each device type as specified in MIL-S-19500

1 2 Physical dimensions See figures 1 and 2

1 3 Maximum ratings Unless otherwise specified, maximum ratings apply to all case outlines and T_A = +25°C

1 3 1 Infrared-emitting diode maximum ratings

V_R	I_F 1/	I_P 2/
V dc	mA dc	A (pk)
2	40	1

1/ Derate linearly to +125°C at 0 67 mA/°C above +65°C
2/ 1 0 µs pulse width, 300 pps

1 3 2 Phototransistor maximum ratings

V_{CEO}	V_{CBO}	V_{EBO}	P_T 1/	I_C
V dc	V dc	V dc	mW	mA dc
35	35	4	300	50

1/ Derate linearly to +125°C at 3 mA/°C above +25°C

1 3 1 Total device ratings

T_J and T_{STG}	V_{IO}
°C	V dc
-65 to +125	1,000 max

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, Defense Electronic Supply Center, ATTN DESC-ELDT, 1507 Wilmington Pike, Dayton, OH 45444-5765 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

AMSC N/A FSC 5980

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1.4 Primary electrical characteristics Unless otherwise specified, electrical characteristics apply to all case outlines and T_A = +25°C.

1 4 1 LED (input) characteristics

Limits	I_R V_R = 2 V dc	V_F I_F = 10 mA dc
	µA dc	V dc
Min		0.8
Max	100	1.3

1.4.2 Phototransistor (output) characteristics.

Limits	$I_{C(OFF)}$ V_{CE} = 20 V dc	$V_{(BR)CEO}$ I_C = 1 mA dc	$V_{(BR)CBO}$ I_C = 100 µA dc	$V_{(BR)CBO}$ I_E = 100 µA dc	h_{FE} V_{CE} = 20 V dc I_C = 10 mA dc		
					4N22 4N22A	4N23 4N23A	4N24 4N24A
	nA	V dc	V dc	V dc			
Min		35	35	4	200	300	400
Max	100						

1.4.2 Coupled (transfer) characteristics.

Limits	R_{IO} at 1,000 V dc see 4.5.4	C_{IO} V = 0 f = 1 MHz see 4 5 4	Phototransistor mode switching (see figure 3) I_C = 100 µA dc					
			4N22	4N22A	4N23	4N23A	4N24	4N24A
			t_r	t_f	t_r	t_f	t_r	t_f
	ohms	pF	µs	µs	µs	µs	µs	µs
Min	10^{11}							
Max		5	15	15	15	15	20	20

Limits	$V_{CE(SAT)}$ I_F = 20 mA			$IC_{(ON)}2$ I_F = 10 mA V_{CE} = 5 V dc			$IC_{(ON)}1$ I_F = 2 mA V_{CE} = 5 V dc		
	I_C = 2.5 mA	I_C = 5 mA	I_C = 10 mA						
	4N22 4N22A	4N23 4N23A	4N24 4N24A	4N22 4N22A	4N23 4N23A	4N24 4N24A	4N22 4N22A	4N23 4N23A	4N24 4N24A
	V dc	V dc	V dc	mA dc	mA dc	mA dc	mA dc	mA dc	mA dc
Min				2.5	6.0	10.0	0.15	0.2	0.4
Max	0.3	0.3	0.3						

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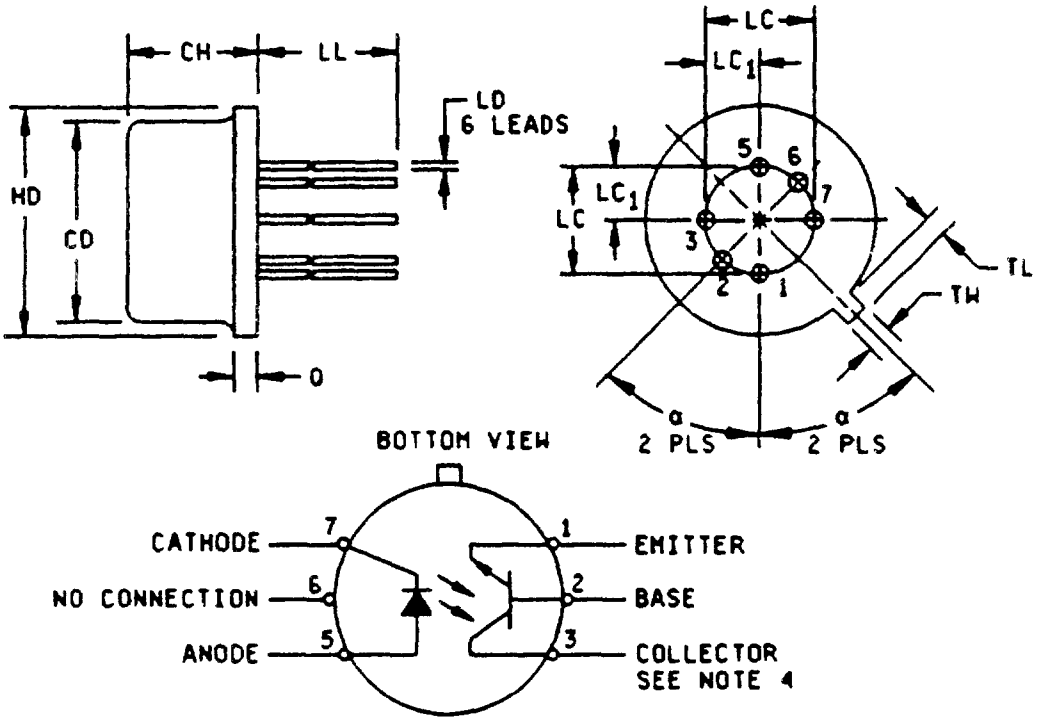


FIGURE 1 Physical dimensions and configuration.

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Dimensions					
Ltr	Inches		Millimeters		Notes
	Min	Max	Min	Max	
HD	.335	.370	8.51	9 40	5
CD	.305	.335	7.75	8.51	5
LD	.016	019	0.41	0.48	5
∝	45° T.P		45° T.P.		3
Q		.040		1.02	
CH	.155	.185	3.94	4.70	
LC	.200 T.P.		5.08 T P.		3
LC_1	.100 T P.		2.54 T.P.		3
TW	.028	.034	0 71	0.86	
TL	.029	.045	0 74	1.14	
LL	.500	.600	12.70	15.24	

NOTES:
1. Dimensions are in inches
2. Metric equivalents are given for general information only
3. T.P. designates true position Leads having maximum diameter .019 inches (0 48 mm) measured in gauging plane .054 + .001, -.000 inch (1.37 +0.03, -0.00 mm) below the seating plane of the device shall be within .007 inch (0.18 mm) of their true position relative to a maximum width tab.
4. 4N22A, 4N23A, and 4N24A have collector isolated from the case.
5. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions and configurations - Continued

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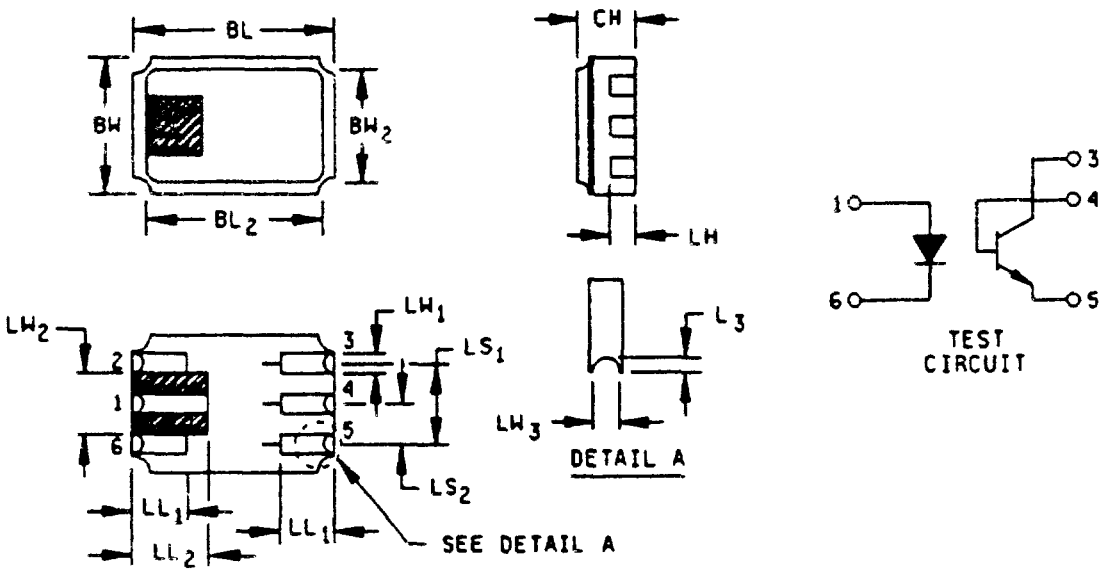


FIGURE 2 Physical dimensions and configuration, surface mounted devices.

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Dimensions					
Ltr	Inches		Millimeters		Notes
	Min	Max	Min	Max	
CH	.066	.080	1 68	2.03	
LH	.026	.034	0.66	0.86	
LW_1	.022	.028	0.56	0.71	
LW_2	.072 REF		1 83 REF		
LW_3	.006	.022	0.15	0.56	
BW	.165	.175	4 19	4.44	
LS_1	.095	.105	2 41	2 67	
LS_2	.045	055	1.14	1.39	
BW_2		.175		4.44	
BL	.240	.250	6.10	6.35	
BL_2		.250		6.35	
LL_1	.060	.070	1.52	1.78	
LL_2	.082	.098	2 08	2.49	
L_3	.003	007	0 08	0 18	
N	6				
ND	3				
NE	0				

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Symbol "N" is the maximum number of terminals. Symbols "ND" and "NE" are number of terminals along the length "BW" and "BL" respectively
4. Pin number designation:
   1 - Anode.
   2 - N/C.
   3 - Collector.
   4 - Base.
   5 - Emitter.
   6 - Cathode.

FIGURE 2 Physical dimensions and configurations, surface mounted devices - Continued.

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2 APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards, and handbooks The following specifications, standards, and handbooks form a part of this specification to the extent specified herein Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2)

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

MIL-STD-883 - Test Methods and Procedures for Microelectronics

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2 2 Non-Government publications. The following document forms a part of this document to the extent specified herein Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Y14.5M - Dimensioning and Tolerancing.

(Application for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036

2.3 Order of precedence In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained

3 REQUIREMENTS.

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500 and as follows:

I_P - - - - - - - - - - - - - - - - - - Peak forward pulse current.

$I_{C(ON)}$ - - - - - - - - - - - - - - - - On-state collector current.

$I_{C(OFF)}$ - - - - - - - - - - - - - - - Off-state collector current.

T_T - - - - - - - - - - - - - - - - - Temperature, terminal pad

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and on figures 1 and 2 herein. With the approval of the qualifying activity, this device is authorized to contain organic material for JANS quality devices

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3 3 1 Lead material and finish Lead material shall be Koval, Alloy 52 or approved equivalent Lead finish shall be gold plated, tin plated, or solder dipped Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.3.2 For eutectic alloyed die of less than 1.5 x 10^{-5} square inches (0.10 mm^2) of surface area. Die shear process control shall be in accordance with the requirements for small area die in MIL-STD-883, method 2019.

3 3.3 JANTX and JANTXV only. Die shear process control on epoxy mounted die is not required.

3.4 Marking Marking shall be in accordance with MIL-S-19500.

3 4 1 Marking for surface mount devices Marking shall be in accordance with MIL-S-19500

3.5 Moisture content. The internal moisture content of the device package shall not exceed 10,000 PPM at +100°C.

4 QUALITY ASSURANCE PROVISIONS

4 1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500 and as specified herein.

4 2 Qualification inspection Qualification inspection shall be in accordance with MIL-S-19500

4.3 Screening (JANTX, JANTXV, and JANS levels). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3	See 4.3.1	See 4.3.1
9 1/	$I_{C(OFF)}$, h_{FE}	$I_{C(OFF)}$, 100 percent read and record.
10	See 4.3.2	See 4.3.2
11 2/	I_R, $I_{C(OFF)}1$, h_{FE} and $I_{C(ON)}2$; $\Delta I_{C(OFF)}1$ = 100 percent of initial value or 25 nA dc, whichever is greater. Δh_{FE} = ±20 percent of initial reading.	I_R, $I_{C(OFF)}1$, h_{FE} and $I_{C(ON)}2$; $\Delta I_{C(OFF)}1$ = 100 percent of initial value or 25 nA dc, whichever is greater.
12	See 4.3.3	See 4.3.3
13	Subgroup 2 of table I herein; $\Delta I_{C(OFF)}1$ = 100 percent of initial value or 25 nA dc, whichever is greater. Δh_{FE} = ±20 percent of initial reading $I_{C(ON)}$ = ±25 percent of initial reading, ΔI_R = 100 percent of initial value or 25 µA dc, whichever is greater.	Subgroup 2 of table I herein; $\Delta I_{C(OFF)}1$ = 100 percent of initial value or 25 nA dc, whichever is greater. Δh_{FE} = ±20 percent of initial reading. $I_{C(ON)}$ = ±25 percent of initial reading; ΔI_R = 100 percent of initial value or 25 µA dc, whichever is greater.

1/ Screen 9 may be omitted if $I_{C(OFF)}1$ in screen 11 is determined using the maximum limit of $I_{C(OFF)}1 \leq 25$ nA (see screen 11)

2/ If screen 9 was omitted, $I_{C(OFF)}1$ maximum limit is 25 nA and $\Delta I_{C(OFF)}1$ is not required

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4 3 1 Thermal shock (temperature cycling) All devices shall be subjected to thermal shock (temperature cycling) in accordance with MIL-STD-750, method 1051, test condition B, except $T_{(min)}$ = -55°C; 10 cycles, time at temperature extremes = 15 minutes minimum

4.3 1.1 Monitored thermal shock (temperature cycling) (JANS only). One cycle on monitored cycling shall be performed on 100 percent of the devices. This test shall be performed any time after the completion of the thermal shock test specified or it may be the last of the 10 thermal cycles. All junctions shall be monitored for electrical continuity. Any discontinuity shall be cause for rejection of the device(s) under test. If 10 percent or more of the number of devices subjected to monitored temperature cycling fail, the entire lot shall be rejected as JANS types.

4.3.2 High temperature reverse bias (HTRB). All devices shall be subjected to high temperature reverse bias in accordance with MIL-STD-750, method 1039, test condition A, T_A = +125°C, I_F = 0, V_{CB} = 28 V dc for 48 hours minimum.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: Burn-in circuit (see figure 4); V_{CC} = 20 V dc, V_{CE} = 10 V dc ±5 V dc, I_F = 40 mA dc, P_T = 275 ±25 mW. For surface mount devices, T_A = room ambient as defined in the general requirements of 4.5 of MIL-STD-750; T_T = +100°C to +125°C for surface mounted devices. No heat sink or forced air cooling on the devices shall be permitted.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4 4 1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.

4.4.2 Group B inspection Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JAN and JANTX) of MIL-S-19500 and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2 1 Group B inspection, table IVa (JANS) of MIL-S-19500.

Subgroup	Method	Conditions
B3	1051	Test condition B.
B4	1037	I_F = 20 mA dc; V_{CE} = 10 V dc; t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. P_T = 275 ±25 mW at T_A = room ambient as defined in the general requirements of MIL-STD-750 No heat sink or forced air cooling shall be permitted on the devices

4.4.2 1 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500

Subgroup	Method	Conditions
B2	1051	Test condition B
B3	1027	I_F = 20 mA dc, V_{CE} = 10 V dc; t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. P_T = 275 ±25 mW at T_A = room ambient as defined in the general requirements of MIL-STD-750. No heat sink or forced air cooling shall be permitted on the devices.

4.4.3 Group C inspection (all quality levels) Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

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4.4.3.1 Group C inspection, table III of MIL-S-19500

Subgroup	Method	Conditions
C2	2036	Test condition E Not applicable to surface mount devices.
C3	2006	Nonoperating, 30,000 G, X_1, Y_1, and Y_2 orientations.
C6	1026	I_F = 20 mA dc; V_{CE} = 10 V dc; t_{on} = t_{off} = 3 minutes minimum for 6,000 cycles P_T = 275 ±25 mW at T_A = room ambient as defined in the general requirements of MIL-STD-750 No heat sink or forced air cooling shall be permitted on the devices.

4.5 Method of inspection Methods of inspection shall be as specified in the appropriate tables and as follows

4.5 1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5 2 Input LED tests These tests shall be performed with the output transistor terminals open.

4 5.3 Output transistor tests These tests shall be performed with the input LED terminals open.

4 5 4 Isolation and coupling capacitance tests. The tests shall be performed between both input terminals 5 and 7 shorted together and the output terminals 1, 2, and 3 shorted together (see figure 1).

4.5.5 Monitored thermal shock (temperature cycling) One cycle of monitored temperature cycling shall be performed. All junctions shall be monitored for electrical continuity Any discontinuity shall be cause for rejection of the device under test.

4.6 Internal visual (PRECAP) inspection (JANTXV level only). Internal visual inspection shall be performed in accordance with MIL-STD-750, methods 2072 and 2073 and as specified herein. Method 2072 shall be used for inspecting transistor die and completed assembly. Method 2073 shall be used for inspecting light emitting diode.

4.6.1 Gallium arsenide or gallium aluminum arsenide light emitting diode die inspection. The die shall be inspected under 100X (minimum).

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TABLE I Group A inspection

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
(LED characteristics)						
Reverse current	4016	V_R = 2 V dc (see 4.5.2)	I_R		100	µA dc
Forward voltage	4011	I_F = 10 mA dc (see 4.5.2)	V_{F1}	0.8	1.3	V dc
(Transistor characteristics)						
Collector to emitter breakdown voltage	3011	I_C = 1.0 mA dc; I_B = 0, (see 4.5.3)	$V_{(BR)CEO}$	35		V dc
Collector to base breakdown voltage	3001	I_C = 100 µA dc, I_F = 0, I_E = 0 (see 4.5.3)	$V_{(BR)CBO}$	35		V dc
Emitter to base breakdown voltage	3026	I_E = 100 µA dc; I_C = 0, I_F = 0 (see 4.5.3)	$V_{(BR)EBO}$	4		V dc
Off-state collector current (phototransistor mode)	3041	Bias condition B; V_{CE} = 20 V dc; I_B = 0, I_F = 0 (see 4.5.3)	$I_{C(OFF)1}$		100	nA dc
Forward-current transfer ratio 2/	3076	V_{CE} = 5.0 V dc; I_C = 10 mA dc, I_F = 0 pulsed (see 4.5.3)	h_{FE}			
4N22, 4N22A				200		
4N23, 4N23A				300		
4N24, 4N24A				400		
(Coupler characteristics)						
On-state collector current 2/ (phototransistor mode)	3041	V_{CE} = 5 V dc; I_F = 2 mA dc	$I_{C(ON)1}$			
4N22, 4N22A				0.15		mA dc
4N23, 4N23A				0.2		mA dc
4N24, 4N24A				0.4		mA dc
On-state collector current 2/ (phototransistor mode)	3041	V_{CE} = 5 V dc; I_F = 10 mA dc	$I_{C(ON)2}$			
4N22, 4N22A				2.5		mA dc
4N23, 4N23A				6.0		mA dc
4N24, 4N24A				10.0		mA dc

See footnotes at end of table

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TABLE I Group A inspection - Continued

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Collector to emitter saturated voltage	3071	I_F = 20 mA dc; Pulsed (see 4.5.1)	$V_{CE(sat)}$		0.3	V dc
4N22, 4N22A 4N23, 4N23A 4N24, 4N24A		I_C = 2.5 mA dc I_C = 5 mA dc I_C = 10 mA dc				
Subgroup 3						
High-temperature operation:		T_A = +100°C				
(Transistor characteristics)						
Off-state collector current (phototransistor mode)	3041	Bias condition B, V_{CE} = 20 V dc; I_B = 0, I_F = 0 (see 4.5.3)	$I_{C(OFF)2}$		100	µA dc
(Coupler characteristics)						
On-state collector current 2/ (phototransistor mode)	3041	V_{CE} = 5 V dc, I_F = 10 mA dc	$I_{C(ON)3}$			
4N22, 4N22A 4N23, 4N23A 4N24, 4N24A				1.0 2.5 4.0		mA dc mA dc mA dc
(LED characteristics)						
Forward voltage	4011	I_F = 10 mA dc (see 4.5.2)	V_{F2}	0.7	1.2	V dc
On-state collector current 2/ (phototransistor mode)	3041	V_{CE} = 5 V dc, I_F = 10 mA dc	$I_{C(ON)4}$			
4N22, 4N22A 4N23, 4N23A 4N24, 4N24A				1.0 2.5 4.0		mA dc mA dc mA dc
(LED characteristics)						
Forward voltage	4011	I_F = 10 mA dc (see 4.5.2)	V_{F2}	1.0	1.5	V dc

See footnote at end of table

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TABLE I. Group A inspection - Continued

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Input to output capacitance		$\|V_{IN} - V_{Out}\| = 0$ (see 4.5.4) f = 1 MHz	C_{IO}		5	pF
Input to output internal resistance		$\|V_{IN} - V_{Out}\| = 1$ kV dc (see 4.5.4)	R_{IO}	10^{11}		Ω
Rise time 2/ (phototransistor mode)		V_{CC} = 10 V dc, I_F = 10 mA dc; R_L = 100 Ω (see figure 3)	t_r			
4N22, 4N22A					15	µs
4N23, 4N23A					15	µs
4N24, 4N24A					20	µs
Fall time 2/ (phototransistor mode)		V_{CC} = 10 V dc; I_F = 10 mA dc, R_L = 100 Ω (see figure 3)	t_f			
4N22, 4N22A					15	µs
4N23, 4N23A					15	µs
4N24, 4N24A					20	µs
Subgroups 5 and 6						
Not applicable						
Subgroup 7						
Monitored thermal shock (temperature cycling)		See 4.5.5				

1/ For sampling plan, see MIL-S 19500
2/ Unless otherwise specified, tests and conditions apply to all case outlines

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TABLE II Groups B and C electrical measurements 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1	Off-state collector current	3041	V_{CE} = 20 V dc, I_F = 0, condition D (see 4.5.3)	$I_{C(OFF)1}$		100	nA dc
2	On-state collector current 4/		V_{CE} = 5 V dc, I_F = 10 mA dc	$I_{C(ON)2}$			
	4N22, 4N22A				2.5		mA dc
	4N23, 4N23A				6.0		mA dc
	4N24, 4N24A				10 0		mA dc
3	Collector to emitter saturation voltage		I_F = 20 mA dc	$V_{CE(SAT)}$		0.3	V dc
	4N22, 4N22A		I_C = 2.5 mA dc				
	4N23, 4N23A		I_C = 5 mA dc				
	4N24, 4N24A		I_C = 10 mA dc				
4	Off-state collector current	3041	V_{CE} = 20 V dc I_F = 0, condition D (see 4.5.3)	$\Delta I_{C(OFF)1}$	±100 percent of initial value or 25 nA dc, whichever is greater		
5.	On-state collector current 4/		V_{CE} = 5 V dc, I_F = 10 mA dc	$\Delta I_{C(ON)2}$ 5/	±25 percent change from initial reading.		

1/ The electrical measurements for table IVa (JANS) of MIL-S-19500 are as follows:

a. Subgroup 3, see table II herein, steps 1, 2, and 3.

b. Subgroup 4, see table II herein, steps 1, 2, and 3.

c. Subgroup 5, see table II herein, steps 1, 2, 4, and 5.

2/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows.

a. Subgroup 2, see table II herein, steps 1, 2, and 3

b. Subgroup 3, see table II herein, steps 1, 2, 4, and 5

c. Subgroup 6, see table II herein, steps 4 and 5.

3/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1, 2, and 3 (JANS); and steps 1 and 2 (JANTX and JANTXV).

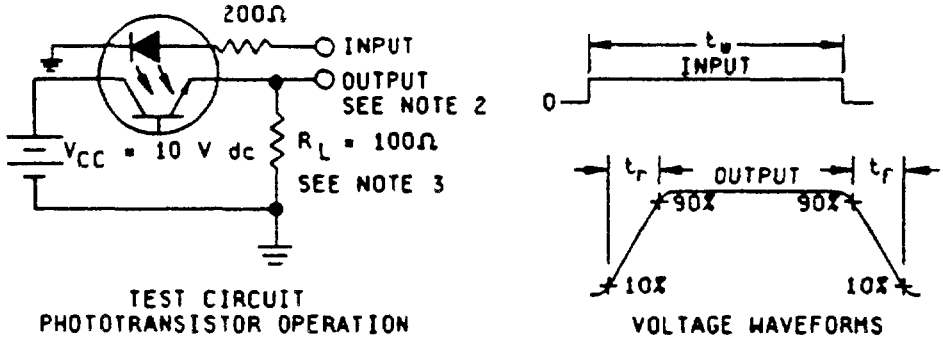
b. Subgroup 3, see table II herein, steps 2 and 3.

c. Subgroup 6, see table II herein, steps 4 and 5.

4/ Unless otherwise specified, tests and conditions apply to all case outlines

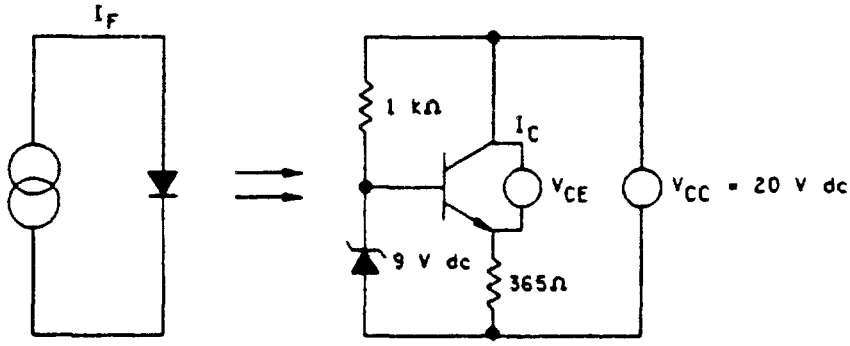
5/ Devices which exceed the group A limits for this test shall not be acceptable for shipment.

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NOTES:
1. The input waveform is supplied by a generator with the following characteristics. Z_{OUT} = 50 Ω, $t_r \leq 15$ ns, duty cycle = 1 percent, t_w = 100 μs
2. The output waveform is monitored on an oscilloscope with the following characteristics: $t_r \leq 1$ ns, R_{IN} = 1 MΩ, C_{IN} = 20 pF.
3. Adjust amplitude of input for: I_F = 10 mA dc.

FIGURE 3 Switching times



NOTES:
1. I_F = 20 mA dc for operation life tests.
2. I_F = 40 mA dc for burn-in of TX and TXV

FIGURE 4 Operation life test burn-in circuit

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5 PACKAGING

5.1 Packaging requirements The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following

a Issue of DODISS to be cited in the solicitation.

b Lead finish as specified (see 3.3.1).

c Type designation and product assurance level.

6.3 Changes from previous issue Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes

CONCLUDING MATERIAL

Custodians:
Air Force 17
NASA - NA

Review activities:
Air Force - 19, 85, 99
DLA - ES

Preparing activity:
NASA - NA

Agent:
DLA - ES

(Project 5980-0012)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1 The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given

2 The submitter of this form must complete blocks 4, 5, 6, and 7

3 The preparing activity must provide a reply within 30 days from receipt of the form

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/486D	2. DOCUMENT DATE (YYMMDD) 950627

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, COUPLER, OPTOELECTRIC SOLID STATE, TYPES 4N22, 4N22A, 4N23, 4N23A, 4N24, 4N24A, 4N22U, 4N22AU, 4N23U, 4N23AU, 4N24U, 4N24AU, JAN, JANTX, JANTXV, AND JANS

4 NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed)

5 REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, first, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

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c ADDRESS (Include Zip Code) Defense Electronic Supply Center ATTN DESC-ELDT Dayton, OH 45444-5765	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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